SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA1714

COURSE NAME: Artificial Intelligence

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks:30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I V Murale Karthik *(Name / Reg No)* (192312567) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	18
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	18
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps, multiple errors	No clear process	23
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors, incorrect final answer	No valid calculations	18
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	13

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1) Premium Subscription Access:-

Step 1:- Proportional logic Representation

- P = "Amitha pays subscription fee"
- Q = "Anita gets premium access"

Knowledge Base:-

- Rule 1: $P \rightarrow Q$
- Fact 2: P

Goal: Prove Q

Step 2:- Convert to CNF & Negate Goal:-

Rule 1: $P \rightarrow Q = \neg P \vee Q$ (CNF)

Fact 2: P (CNF)

Negated Goal: $\neg Q$ (CNF)

Clauses.

Clause	Formula
C1	$\neg P \vee Q$
C2	P
C3	$\neg Q$

Step 3:-

C_1: $\neg P \vee Q$

C_2 P

C_4: Q

C_4: Q

C_3: $\neg Q$

C_5: $\square$ [Empty clause]

Step 4:- Empty clause Derived → Goal proved.

2) Smart Device Connectivity:

Step 1:- W = "Smart TV is connected to WiFi"

I = "Smart TV can access the internet".

Knowledge Base:-

- Rule 1 : $W \rightarrow I$
- Fact 2: W

Step 2:-

Clauses

Clause	Formula
C_1	$\neg W \vee I$
C_2	W
C_3	$\neg I$

Step 3:- Resolve C_1 & C_2

C_1 : $\neg W \vee I$

C_2 : W

C_4 : I

Resolve C_4 & C_3

C_4 : I

C_3 : $\neg I$

C_5 : $\square$ [Empty clause]

Step 4:-

Empty clause Derived => Goal proved

$\therefore$ Smart TV can access the internet.

3) Machine Auto Shutdown:-

Sol: Step 1:- O = "Machine Overheats"

S = "Machine shuts down automatically."

Knowledge Base:-

Rule 1:- $O \rightarrow S$

Fact 2: O

Goal: Prove S

Step 2:-

Clauses:-

Clause	Formula
C_1	$\neg O \vee S$
C_2	O
C_3	$\neg S$

Step 3:- Resolve C_1 & C_2

C_1 :- $\neg O \vee S$

C_2 :- O

C_4 :- S

Resolve C_4 & C_3

C_4 : S

C_3 : $\neg S$

C_5 : $\square$ [Empty clause]

Step 4:-

Empty clause derived $\rightarrow$ Goal proved

$\therefore$ Machine shuts down automatically

4) Account Verification:-

Sol: Step 1:- U = "User uploads valid ID"

V = "Account is verified".

Knowledge base:-

Rule 1 : $U \rightarrow V$

Fact 2 : U.

Goal:- Prove V

Step 2:-

Clauses

Clause	Formula
C_1	$\neg U \vee V$
C_2	U
C_3	$\neg V$

Step 3:-

Resolve C_1 & C_2

$C_1: \neg U \vee V$

$C_2: U$

$C_4: V$.

Resolve C_4 & C_3

$C_4: V$

$C_3: \neg V$

$C_5: \square$ [Empty clause]

Step 4:- Empty clause Derived -> Goal proved.

∴ Account is verified

5) Employee Benefits Eligibility:-

Sol: Step 1:- P = "Employee finishes probation"

S = "Employee becomes permanent staff

B = "Employee gets full benefits")

Knowledge Base:-

Rule 1:- $P \to S$

Rule 2: $S \to B$

Fact 3: P.

Step 2:-

Resolve C_1 & C_3

C_1 $\neg P \vee S$

C_3 P

C_5: S

Resolve C_5 & C_2

C_5: S

C_2: $\neg S \vee B$

C_6 B

Resolve C_6 & C_4

C_6 : B,

C_4 : $\neg B$

C_7 : $\square$ [Empty clause]

Hence proved.